

YANFEI DAI

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Education

Cornell University

Master of Computing and Information Science

Aug. 2023 – Dec. 2024

Ithaca, New York

Nanjing University

Bachelor of Engineering

Sep. 2018 – Jun. 2023

Nanjing, Jiangsu

Experience

PayPal Inc.

Machine Learning Engineer

Feb. 2025 – Present

San Jose, CA

- Designed and ran ablation studies across 10+ **two-tower (dual-encoder) retrieval** variants (DNN, **Deep & Cross Network / DCN-v2**), isolating how **sequence aggregation**, **Matryoshka embeddings**, and **BERT-based item features** affect **recall@K** — lifting **recall@100** by **~3%** over the production baseline.
- Cut candidate-retrieval latency by **~25%** (p99 ~90ms to ~68ms) on a **~10M-item** corpus by integrating **FAISS approximate nearest-neighbor (ANN)** search, holding **recall** within **~2%** of exact (brute-force) search.
- Built and deployed an **agentic AI** fraud-analysis workflow in which a **planning agent** orchestrates specialized sub-agents through **MCP (Model Context Protocol)** tool calls over a **retrieval (RAG)** layer to automate first-pass case triage; owned the **CI/CD** and evaluation harness, trimming repetitive analyst review by **~30%**.
- Trained lightweight **LightGBM** models to complement rule-based fraud defenses on emerging patterns where production models lagged, and surfaced top risk drivers via **feature-importance** analysis for the strategy team — improving targeted rule precision by **~5%**.

ProtagoLabs / Netmind.AI

Machine Learning Engineer

May. 2024 – Dec. 2024

Vienna, VA

- Optimized **LLM serving / inference** for NetMind.AI's token-metered model-serving platform across owned and rented **GPU** infrastructure, supporting a **10,000+** daily-active-user base on NetMind Power.
- Applied **quantization (TensorRT)** with multi-GPU / multi-node serving via **DeepSpeed**, improving throughput **~3x** over the FP16 baseline with benchmark accuracy loss under **0.05**.
- Added a self-service **fine-tuning** workflow to the platform using **PEFT (LoRA)** and **SGLang**, letting end-users customize models through a one-click, click-to-tune flow.
- Worked with research on **post-training** experiments — **CPT** for long-form reasoning, **SFT** for multi-turn dialogue, and **RL** with heuristic rewards — helping surface long-output and **reward-hacking** failure modes that informed later iterations.

Boston Derm Advocate

Machine Learning Engineer Intern

Nov. 2023 – May. 2024

Remote

- Designed and prototyped a personalized **skincare recommendation system** using a **two-tower (dual-encoder) retrieval** architecture that learns joint user-product **embeddings** to rank products by predicted preference.
- Built the **feature-engineering** and data-preparation pipeline from user skin-type surveys and product ingredient metadata, adding a **content-based fallback** to cover **cold-start** users with sparse history.
- Trained and tuned gradient-boosted models (**XGBoost**) for preference prediction, using systematic **feature ablation** and metric-driven selection (**F1, AUC**) to finalize the model — improving offline relevance by **~5%** over a popularity baseline.
- Stood up a reproducible **offline evaluation** workflow (**recall@K, precision@K**) to compare candidate models consistently, and packaged the model and results for handoff to the founding engineering team.

Technical Skills

Recommendation / Retrieval: two-tower / dual-encoder retrieval, candidate retrieval, ranking, CTR / CVR prediction, learning-to-rank, Deep & Cross Network (DCN), DLRM, embeddings, approximate nearest neighbor (ANN), FAISS, HNSW, vector search, A/B testing, recall@K, NDCG, AUC

LLM / Agentic AI: agentic AI, multi-agent orchestration, tool use / function calling, MCP (Model Context Protocol), RAG (Retrieval-Augmented Generation), fine-tuning, PEFT, LoRA, SFT, RLHF, post-training, prompt engineering, evals

ML Systems / Serving: LLM serving, inference optimization, quantization, TensorRT, DeepSpeed, SGLang, vLLM, distributed / multi-GPU training, latency & throughput optimization

Frameworks & Tools: Python, PyTorch, TensorFlow, LightGBM, XGBoost, Spark, SQL, feature engineering, Git, Docker, CI/CD